



2019 SIOP MACHINE LEARNING COMPETITION

Predicting Personality from Text

Estimating the Big Five from five open-ended answers — a leakage-safe, LLM-powered stacked solution

PRIVATE-TEST MEAN r

0.3215

across the five Big Five traits



WHAT WE PREDICT

The Task

Each respondent answers **five open-ended situational-judgment items**. From that text alone, predict their five **Big Five trait scores** (A · C · E · N · O), scored by **mean Pearson r** — so only rank order matters.

1,088

TRAIN

300

DEV

300

TEST



VERIFIED, LEAKAGE-SAFE

Results

0.3635

TRAIN-CV (OOF)

0.360

PUBLIC · DEV

0.3215

PRIVATE · TEST

PER-TRAIT (PRIVATE TEST)

E

0.41

A

0.39

C

0.28

O

0.27

N

0.25



LEAKAGE-SAFE STACKED GENERALIZATION

The Approach



LLM bases

Claude · zero-shot

- Haiku 4.5 scorer ×4 prompts
- Sonnet 4.6 second judge
- 16-dim behavioral subfeatures
- Role-play BFI-2 questionnaire



Classical bases

proven pre-LLM signal

- e5-large per-prompt SVR (RBF)
- e5-large all-text SVR
- TF-IDF char + word Ridge
- Engineered psycholinguistic

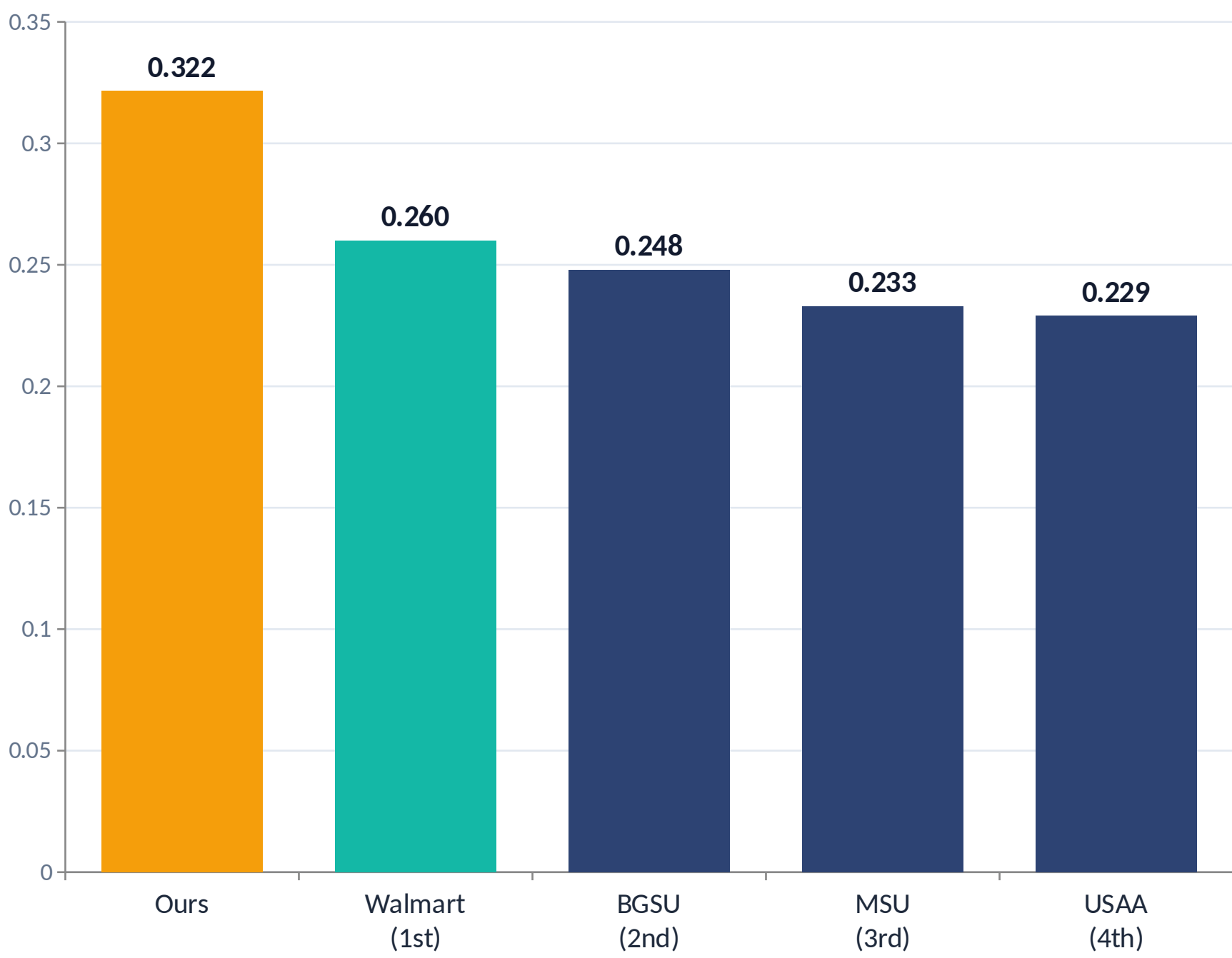


Per-trait Ridge meta-learner — fit on Train out-of-fold only



PRIVATE TEST · VS THE 2019 TOP-4

How It Compares



THREE LEVERS NO 2019 TEAM HAD

What Set It Apart



Frontier LLMs as zero-shot extractors

Instruction-tuned Claude models bring a vast prior no 1,088-row model could learn — the single biggest gain.



Multi-prompt + second judge

Four prompt framings and an independent Sonnet judge turn that prior into a low-variance signal.



Role-play questionnaire simulation

Haiku answers a 30-item BFI-2 battery in the respondent's persona; reverse-scored to trait estimates.



BUILT TO GENERALIZE

Why the number is trustworthy



Fit only on Train (n=1,088) — no estimator sees Dev/Test labels



Select on Train-CV (out-of-fold) + Dev; touch the private Test once



Zero-shot LLM outputs use no labels — they cannot leak across splits



Independently re-scored against ground truth: Test 0.3215, Dev 0.360



Code, write-up & reproduction: github.com/mjmonnot/siop-2019-personality-winning-solution

2019 SIOP Machine Learning Competition · Big Five from open-ended text